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Paleomagnetic and geochronological study of a geomagnetic field reversal or excursion recorded in pliocene volcanic rocks from Georgia (Lesser Caucasus)

Pierre Camps [a,*], Gilles Ruffet [b], Valera P. Shcherbakov [c], Valja V. Shcherbakova [c], Michel Prévot [a], Alexis Moussine-Pouchkine [d], Lolij Sholpo [e], Afthandil Goguitchaichvili [f], Bejan Asanidzé [g]

[a] *Laboratoire de tectonique et géophysique, Unité associée au CNRS Case 060, Université de Montpellier II, Place Eugène Bataillon, 34095 Montpellier Cedex 5, France*

[b] *Institut de Géodynamique, Université de Nice, Nice, France*

[c] *Institute of Physics of the Earth, Moscow, Russia*

[d] *Géofluides Bassin Eau, Université de Montpellier II, Montpellier, France*

[e] *Institute of Terrestrial Magnetism, Saint Petersburg, Russia*

[f] *Geophysical Department, University of Tbilissi, Tbilissi, Georgia*

[g] *Institute of Caucasian Mineral Resource, Georgia*

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Abstract

A 250-m thick volcanic sequence of some 63 Pliocene lava flows from Georgia (Akhalkalaki plateau) which was previously thought to record the Gauss–Matuyama reversal was re-studied in detail. About 400 oriented cores were sampled along four sub-vertical sections partially overlapping each other. At the base of the section, 16 consecutive lava flows were found to record intermediate directions. The overlying flows are all reversely magnetized. With the exception of the upper part of the sequence, four groups of three to 11 consecutive lava flows with strictly the same paleodirection can be recognized. The location of the intermediate virtual geomagnetic poles (VGPs) does not correspond to that of the VGP clusters previously proposed as indicating a long lasting near dipole configuration during reversals. The field paleointensity is close to one fifth of the present field at the bottom of the section and tends to increase uphill. Age determination ($^{40}Ar/^{39}Ar$) provides an estimate of 3.6 ± 0.06 Ma, which leaves us with two possible interpretations: the transitional vectors could correspond either to an excursion within chron 2Ar or to the upper (N–R) Cochiti–Gilbert reversal.

1. Introduction

The morphology of the geomagnetic field during reversals and excursions is a major problem in geomagnetism. Laj et al.'s (1991, 1992) compilations of transitional virtual geomagnetic poles (VGP) recorded by sedimentary rocks support the existence of two opposite preferred longitude sectors. However, Quidelleur et al. (1994) showed that a similar grouping can be observed if VGPs are plotted with the longitude of the corresponding site of observation taken as origin. Thus, these preferred sectors might be due to the combination of some artefacts in

* Corresponding author.

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the field recording (Rochette, 1990) with the quite irregular longitudinal distribution of the sites of observation (Courtillot and Valet, 1995). If one turns to volcanic rocks, which are more reliable field recorders, the most extensive compilation carried out so far indicates that the longitudinal distribution of VGPs is uniform (Prévot and Camps, 1993). Yet, the geographical distribution of the sites of observations is poor. There are large areas at the Earth's surface like continental Eurasia or most of the Southern hemisphere where almost no reliable observations are available regarding transitional fields. Precise paleomagnetic data from reversals or excursions recorded in these areas, like the data recently obtained from Chile by Brown et al. (1994), are strongly needed. Moreover, geochronological dating is necessary for a proper identification of each reversal. This is a prerequisite for being able in the near future to investigate the morphology of individual reversals rather than only their statistical characteristics. With this objective in mind, we carried out an extensive paleomagnetic and geochronologic study of basaltic flows from South Georgia (Lesser Caucasus) having previously been reported to record the Gauss–Matuyama transition (Borisova and Sologashvili, 1985).

2. Geology and sampling

An important volcanism occurred in South Georgia from Neogene to Anthropogene. The sequence sampled is part of the Akhalkalaki plateau. The Akhalkalaki formation consists mainly of doleritic basalts or, exceptionally, of andesites. Most of the volcanic activity is assigned to Pliocene but has been continuing during Quaternary.

The main section sampled is situated at 41°28′37″

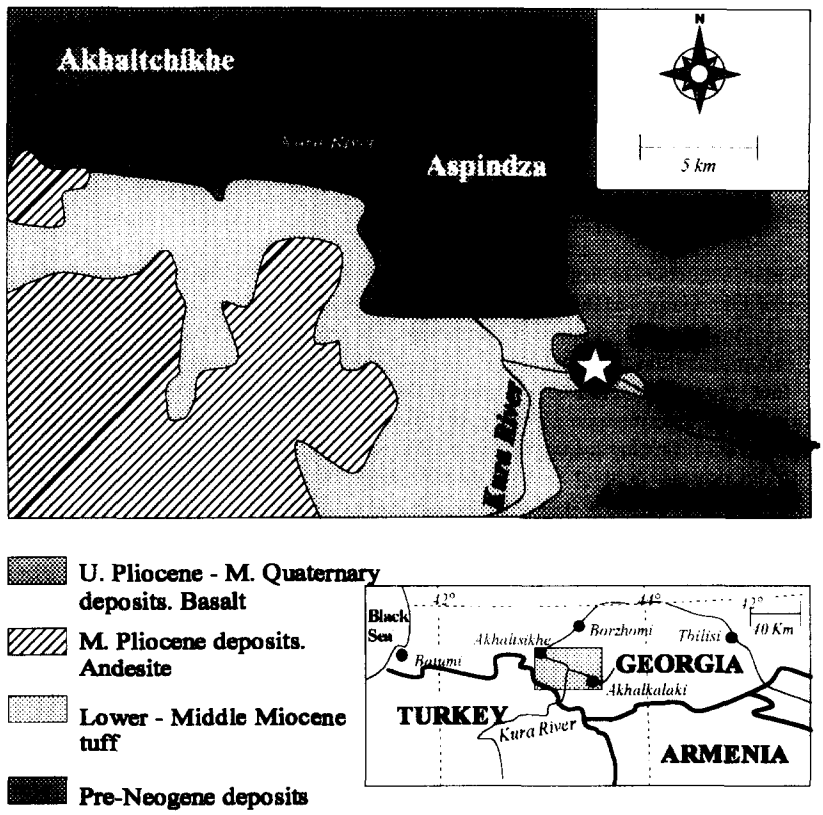


Fig. 1. Schematic geologic map showing location of the sampling site.

N latitude and 43°22′51″ E longitude. That is 1 km S.SE of the village of Thoki, west of the Paravani and Tchobareti brook confluent, near the road from Aspindza to Akhalkalaki (Fig. 1). At the bottom of this section, the Akhalkalaki formation, made of sub-alcaline intraplate basalts, lies discordantly on the Goderzi formation made of volcanic tuffs reported of lower middle Miocene age. An erosion surface visible at the top of this tuff suite forms paleovalleys, later filled with the basaltic flows of the Akhalkalaki formation. This formation consists of two parts, separated by a second erosion surface. The lower Akhalkalaki sequence is approximately 250 m thick. It consists of some 24 gray, massive and coarse-grained lava flows with a maximum thickness of 30 m. Several of these flows show evidence of weathering, with however relatively fresh rock near the center of preserved bowls. Sologashvili (1986) reported that the lower part of this sequence provides intermediate magnetization directions while the upper part is reversely magnetized. The upper Akhalkalaki sequence constitutes the upper part of the Thoki plateau and is approximately 150 m thick. The lava flows are fresh columnar dark gray fine-grained basalts only 2–3 m thick in general. They are reversely magnetized (Sologashvili, 1986). Due to the time needed for erosion, there is no doubt that some period of quiescence occurred between the emissions of the lower and upper Akhalkalaki volcanic sequences.

The main section sampled (called W) is situated along a crest overlooking the road. This section starts in a paleotalweg cut within the Goderzi formation and ends at the second erosion surface. Nineteen consecutive flows were sampled between the two erosion surfaces. Section Y, located 400 m north of section W, corresponds also to the lower Akhalkalaki sequence. Eighteen flows were sampled, the bottom one (1Y) corresponding probably to flow 7W (Fig. 2). Along section Y the contact between the lower and the upper Akhalkalaki sequences is probably somewhat above the uppermost flow sampled (18Y). Section Z, which only covers three flows from the lower Akhalkalaki sequence, is situated 50 m west of

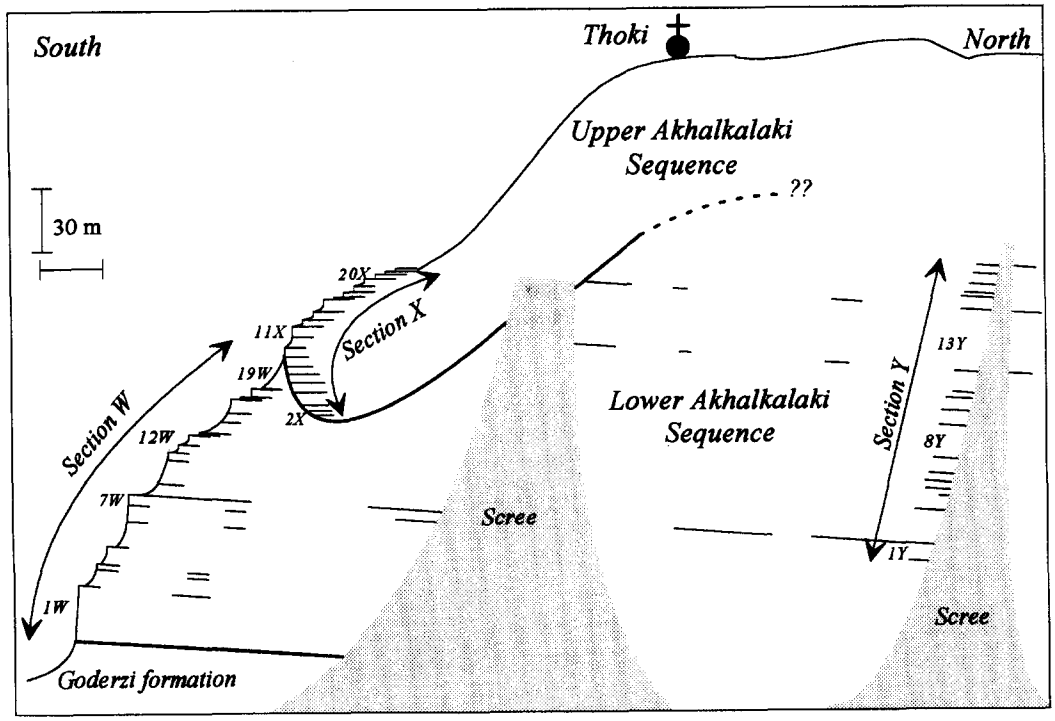


Fig. 2. Schematic cross-section showing location of sampled sections. Arrows denote the approximate location of sections W, X, and Y. Section Z is behind the crest, 50 m west of section W.

section W. The boundaries of these flows are in general perfectly visible and identifiable. However, it is difficult to be sure of the identification of individual flows from one section to the other because of the possible occurrence of vertical faults or lateral pinching out of the flows. All this makes it difficult to establish a one to one stratigraphic correlation between the flows from the different sections. So, our sampling strategy was also guided by measurements using a portable fluxgate probe. A fourth section, called X, corresponds to the lower part of the upper Akhalkalaki sequence. Twenty consecutive

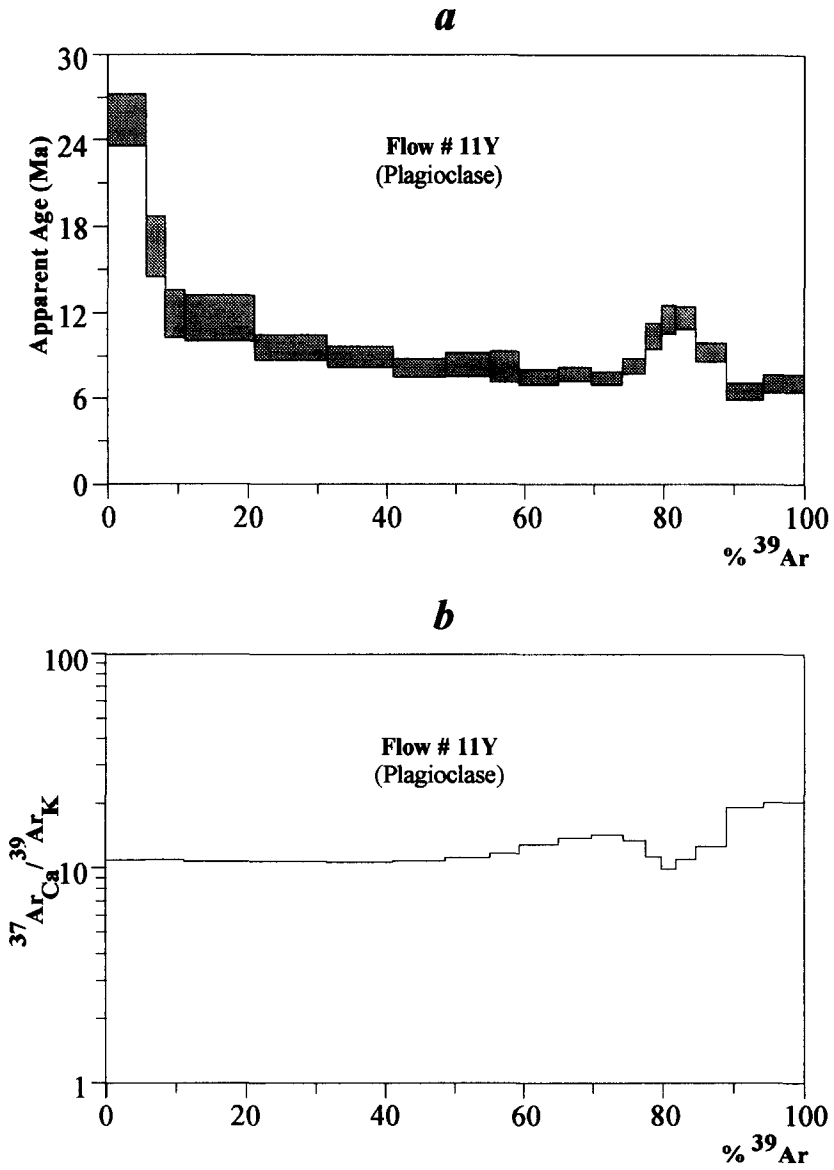


Fig. 3. Plagioclase bulk samples THOKI 11Y. (a) $^{40}Ar/^{39}Ar$ age spectrum. The age error bars for each temperature steps are at the 1σ level. The errors in the J-values are not included. (b) $^{37}Ar_{Ca}/^{39}Ar_{K}$ spectrum.

flows were sampled from the bottom of a paleotalweg uphill (Fig. 2).

Five to ten oriented cores were sampled from each flow using a gasoline portable drill. The lateral extension of the sampling along each section varies from several meters to, more commonly, tens of meters. A total of more than 400 samples was obtained from some 45 distinct flows. The flows show a slight dip (about 4°) towards NE. This dip was supposed to be of topographic origin. Thus, no tectonic correction was applied to the directions of magnetization.

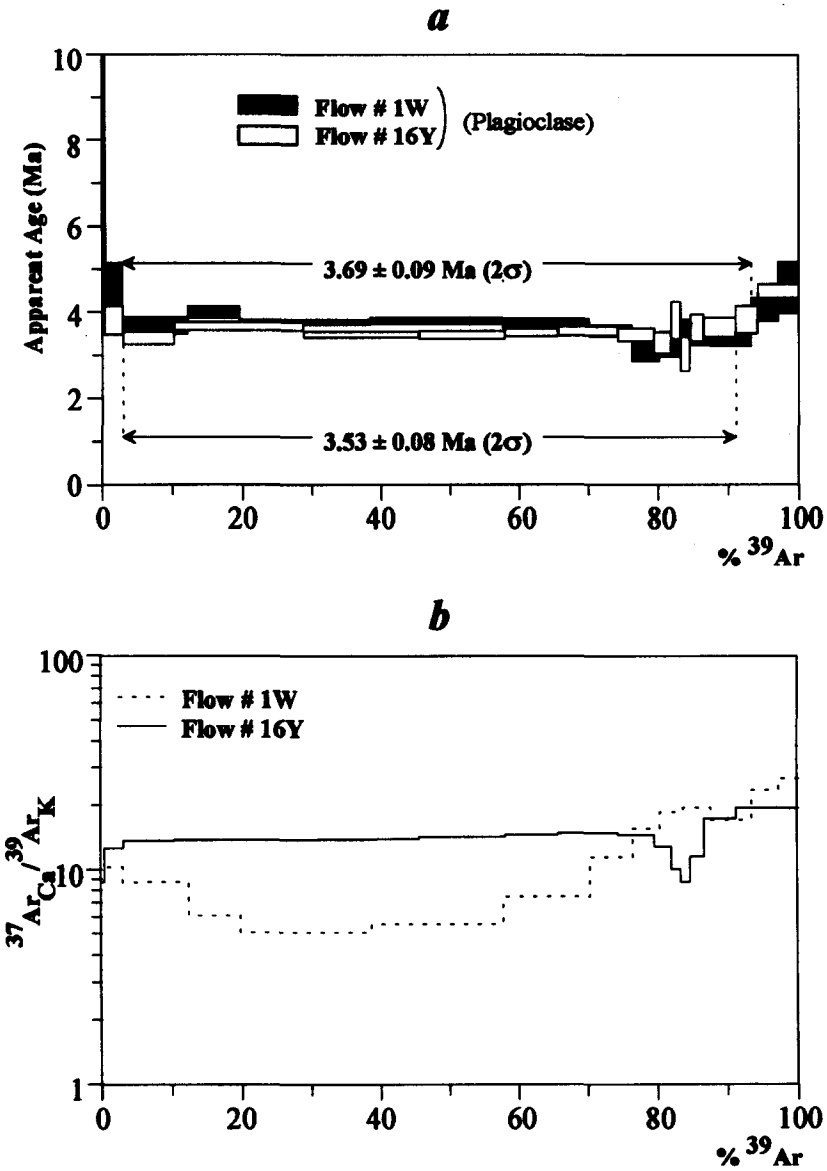


Fig. 4. Plagioclase bulk samples THOKI 1W and THOKI 16Y. (a) $^{40}Ar/^{39}Ar$ age spectra. Same observations as for Fig. 3 (b) $^{37}Ar_{Ca}/^{39}Ar_{K}$.

3. $^{40}AR/^{39}AR$ geochronology

3.1. Analytical procedure

The plagioclase bulk samples (THOKI 1W, 11Y and 16Y) were separated by the classical methods (magnetic separator and heavy liquids). The three final mineral separates (in the range 125–160 μm) were obtained by hand-picking under a binocular microscope. The samples were irradiated in the McMaster reactor (Hamilton, Canada) with a total flux of 6.4×10^{17} n.cm^{-2}. The analyzed samples, wrapped in Al foils, were included in a narrow zone from the same level within the irradiation can to minimize the effects of flux gradient which is estimated, in this zone to be less than $\pm 0.5\%$. The irradiated standard was the biotite Bern 4B (17.25 Ma; Hall et al., 1984, and subsequent analyses performed in the Toronto and Nice laboratories). The step heating procedure is described by Féraud et al (1986).

The mass spectrometer consists of a 120° M.A.S.S.E.® tube, a Bäur Signer® source and an SEV 217® electron-multiplier (total gain: 5×10^{12}) whereas the extraction and purification lines include a Ti-furnace (700°C), a SAES GP50W getter with St101® zirconium–aluminum alloy operating at 400°C and a −95°C cold trap.

Isotopic measurements are corrected for K and Ca isotopic interferences, mass discrimination and atmospheric argon contamination. All errors are quoted at the 1σ level and do not include the errors on the $^{40}Ar^*/^{39}Ar_K$ ratio and age of the monitor. The error in the $^{40}Ar^*/^{39}Ar_K$ ratio of the monitor is included in the plateau age error bars calculation. To define a plateau age, we need at least three consecutive steps, corresponding to a minimum of 70% of the total $^{39}Ar_K$ released, and the individual fraction ages should agree within 2σ with the 'integrated' age of the plateau segment.

3.2. Results

The results are displayed in Fig. 3 and 4. The plagioclase population THOKI 11Y displays an age spectrum (Fig. 3(a)) with a regular decreasing staircase shape, from 25.4 Ma in the low temperature steps down to ca 7 Ma at high temperature, disturbed by a bump between 930°C and 1150°C. The shape of this age spectrum is partly well correlated to the evolution of the $^{37}Ar_{Ca}/^{39}Ar_K$ ratio during the step heating experiment, for the intermediate and high temperatures (Fig. 3(b)).

The plagioclase populations THOKI 1W and 16Y display two age spectra with similar shapes (Fig. 4(a)). Both age spectra show high ages in the first 3% of the $^{39}Ar_K$ released followed by a flat age domain over 80–90% of the total $^{39}Ar_K$ released and finally an increase of the ages for the remaining 10% of the degassing (Fig. 4(b)). The two flat age domains allow the definition of two plateau ages at 3.69 ± 0.04 Ma and 3.53 ± 0.04 Ma for samples THOKI 1W and THOKI 16Y, respectively. Despite similar age spectra, the two samples display distinct $^{37}Ar_{Ca}/^{39}Ar_K$ spectra with a rather flat pattern for sample THOKI 1W and a saddle shape for sample THOKI 16Y.

3.3. Discussion

The high ages in the low temperature steps and the decreasing staircase shape of the THOKI 11Y age spectrum are the most obvious signs of the presence of an excess argon component which probably disturbs the whole experiment and does not allow us to obtain a valid age. The high temperature are bump (Fig. 3(a)) corresponds to a clear decrease of the $^{37}Ar_{Ca}/^{39}Ar_K$ ratio (Fig. 3) which could correspond to the degassing of either an alteration phase, contemporaneous of the cooling of the volcanic flow, or a contamination phase.

The two samples THOKI 1W and 16Y display two 'distinct' plateau ages (Fig. 4a) which agree with their positioning in the volcanic sequence. Nevertheless, as these two ages are concordant at the 2σ level, it is not possible to confirm that the plateau age difference is significant. The validity of the measured ages, disregarding the respect of the plateau age criteria, is confirmed by (1) their concordance and (2) the very distinct shapes of the $^{37}Ar_{Ca}/^{39}Ar_K$ spectra with nevertheless similar age spectra shapes. The saddle shape of the THOKI 1W $^{37}Ar_{Ca}/^{39}Ar_K$ spectrum (Fig. 4(b)) probably corresponds to the simultaneous degassing of the plagioclase and a higher K-content alteration phase, which probably crystallized during the flow cooling.

So we propose that the mean age, 3.60 ± 0.06 Ma (2σ), is the best estimate of the time of emplacement of the lower Akhalkalaki sequence.

4. Rock magnetic properties

4.1. Experimental procedure

Preferably, igneous rocks used for paleodirection and more especially paleointensity determinations must satisfy the following conditions:

1. Be thermally stable at least in the 'active' temperature range where determinations are taken;

2. For paleointensity, the remanence carriers must be fine, i.e. single-domain (SD) or small pseudo-single-domain (PSD) grains. Multi-domain (MD) grains usually give erroneous results (Levi, 1977) in relationship with a violation of the law of independence of partial thermoremanent magnetizations (PTRMs) (Bol'shakov and Shcherbakova, 1979).

3. The part of natural remanent magnetization (NRM) used for paleointensity determinations must be a thermoremanent magnetization (TRM).

4. The magnetic viscosity of the specimens must be small enough to obtain reliable and precise measurements of the remanence during the process of progressive thermal demagnetization. This requirement can be readily fulfilled if the magnetic viscosity index v is lower than 5% (Prévot et al., 1985).

As for the last point, viscosity measurements provided viscosity indexes generally less than 5% except for a few specimens from the very bottom of section W which were excluded from further consideration. To verify the fulfilment of other demands an extensive study of the magnetic properties has been carried out on selected samples:

1. Strong field double heating thermomagnetic curves of induced saturation magnetization (J_s–T). These experiments allowed us to determine the average Curie point of the rock (corresponding to the temperature of the inflection point on J_s–T curve) and check the thermal stability of the ferrimagnetic minerals by comparing the J_s–T curves for the first heating (performed up to 600° to 700°C) and for the second one.

2. Continuous thermal demagnetization of PTRMs. These experiments are crucial to distinguish between MD, PSD or SD carriers of TRM (and NRM as well so far as NRM is indeed a TRM). Bol'shakov and Shcherbakova (1979) were the first to suggest that the domain structure can be inferred from a 'thermomagnetic criterion'. This criterion consists of testing the law of independence of PTRMs which implies that any PTRM(T_1, T_2), in which T_1 is greater than T_2, must be entirely destroyed when heated to T_1. They found that the law is valid for single-domain grains but not for multi-domain ones where a PTRM(T_1, T_2) is only completely demagnetized at temperatures significantly larger than T_1, sometimes reaching the Curie point. For small pseudo-single-domain magnetite grains with sizes lower than 5 μm, the PTRM(T) curves show a pronounced inflection at T_1. Thus such grains can be easily distinguished from single-domain and multi-domain grains.

In addition, magnetic parameters such as the in situ Koenigsberger ratio Q_n, the ratio of saturation remanent magnetization to saturation magnetization (J_{rs}/J_s) and the ratio of remanent coercive force to coercive force (H_{cr}/H_c) were also measured on adjacent unheated specimens.

3. Continuous thermal demagnetization curves of NRM and artificial TRM acquired in a 50 μT field, when the specimen was cooled down from the Curie point after demagnetization of NRM. These experiments were carried out in order to estimate the unblocking temperatures of NRM and to compare NRM and TRM. A similarity between these curves suggests in principle that NRM is actually a TRM provided that no noticeable changes in magnetic mineralogy occurred as a result of heating. Thermomagnetic measurements were carried out using vibrating sample magnetometers with a furnace built in Borok laboratory which allowed to the use of either weak or strong (up to 0.8 T) inducing fields.

One to four specimens from each flow were chosen for each kind of experiment listed above, all heating being performed in air. We studied 51 samples from the lower Akhalkalaki sequence and 18 from the upper sequence. Moreover, several electron microprobe and scanning electron microscope analyses were carried out on samples selected from the paleodirection determinations.

4.2. Results

4.2.1. Nature of magnetic oxides

The Curie temperatures of the basalts from the lower Akhalkalaki sequence lie typically between 550° and 580°C. In a few cases the Curie temperature is higher (600°C) or lower (520°–530°C). We found a single occurrence (flow 19W) of a low-temperature Curie point (212°C), probably indicative of the presence of some Ti-rich titanomagnetite. Hematite is associated with magnetite in flow 14W. In general, the ferrimagnetic mineral appears to be magnetite or Ti-poor titanomagnetite, possibly slightly maghemitized in a few cases. Microprobe analyses confirmed these conclusions. They indicated the presence of inter-growths of Ti-poor titanomagnetite with an abundance of titanohematite lamellae. Such a high-temperature oxidation is usually thought to take place at temperatures higher than 600°C. Thus electron microprobe analyses provide decisive evidence that the primary NRM of these rocks is a TRM. For a few samples, however, no lamellae are visible. In view of the quite high-temperature Curie points of these rocks, close to that of magnetite, it is reasonable to suggest that a similar but finer spinodal decomposition of the original Ti-rich titanomagnetite occurred too. In the absence of any high-temperature exsolution phase which could be identified, we must consider the possibility that this process took place below the Curie point of magnetite (Grommé et al., 1969). If this is the case, then the primary NRM might not be a pure TRM. In contrast, the Curie temperatures of the basalts from the upper Akhalkalaki sequence are constantly low, between 130° and 200°C, which suggests that the remanence carrier is Ti-rich titanomagnetite.

4.2.2. Thermal stability of the ferrimagnetic minerals

For the lower Akhalkalaki sequence, the J_s–T experiments demonstrated a wide variety of degree of thermal stability. Approximately a quarter of the sample indicated small magnetochemical changes (similar shape to the two J_s–T curves and J_s values at room temperature within 10% of each other, Fig. 5(b)). The other samples showed noticeable changes, although not too drastic in many cases (Fig. 5(a)). In contrast, the basalts from the upper Akhalkalaki sequence are extremely irreversible, as could be expected for high-Ti titanomagnetite heated in air. The Curie point increases as a result of heating, which is in agreement with the indications given by the demagnetization curves of NRM (see Fig. 7(e)).

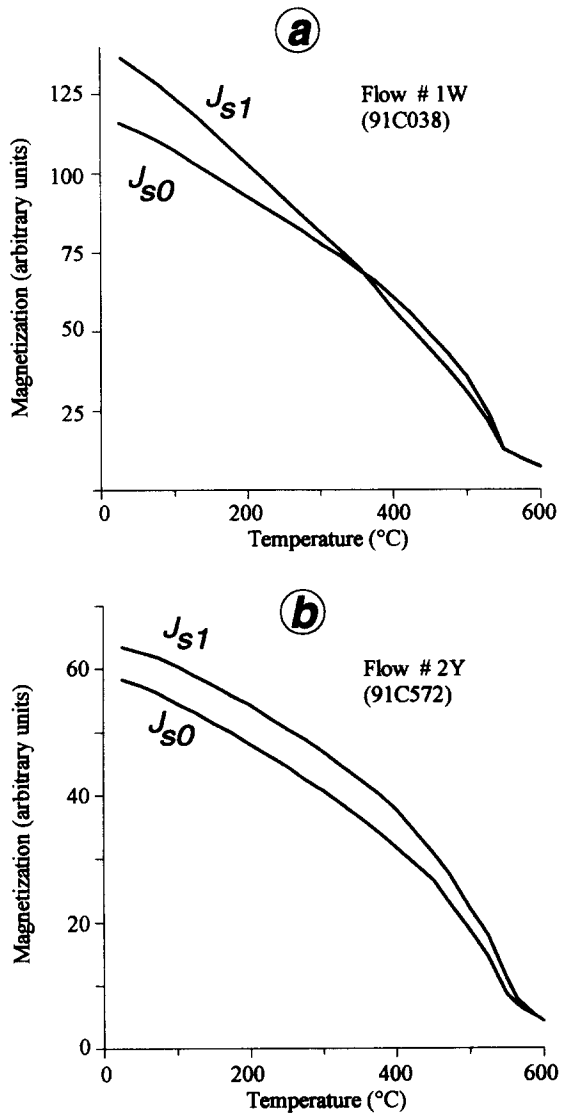


Fig. 5. Types of strong field (0.8 T) double heating thermomagnetic curves of induced saturation magnetization observed for samples of the Thoki basalt. J_{s0} and J_{s1} are the spontaneous magnetizations during the first and second heating respectively. (a) Example of a noticeable magnetochemical change (b) Example of a strong thermal stability.

4.2.3. Domain structure tests

The thermomagnetic criteria for domain structure was applied to PTRMs (300°, 20°C) which were acquired during the cooling down of the sample from the Curie temperature (initially demagnetized state). Both the basalts from the upper and the lower Akhalkalaki sequences were found to behave similarly. A representative example is shown in Fig. 6. We note first that the NRM(T) and TRM(T) curves are relatively close to each other, which indicates the absence of drastic magnetochemical changes due to heating. Obviously, the PTRM(300°, 20°C) is not completely demagnetized at 300°C. The remanence left is slightly less than 25% of the value of the PTRM at room temperature. The PTRM(T) curves die out before reaching the Curie temperature. This behavior is typical of single-domain/pseudo-single-domain mixtures.

Concerning the other magnetic characteristics, the in situ Koenigsberger ratio Q_n was found to vary from 0.5 to 20 in the basalts from the lower sequence, values less than one being observed only in the first flow (1W). Typical values are between 2 and 5. In contrast, the basalts from the upper Akhalkalaki sequence provide Q_n values averaging 50. In both sections the H_{cr}/H_c ratio is about 2 and J_{rs}/J_s vary from 0.05 to 0.25. Altogether, these values are indicative of magnetic grain sizes mostly in the pseudo-single-domain range.

5. Paleodirection determinations

5.1. Experimental procedure

Before magnetic cleaning, the viscosity index v (Thellier and Thellier, 1944) was first determined on the entire collection. This determination included measuring natural remanent magnetization (NRM) after a 3-week storage in zero field. Then, a pilot sample from each flow was thermally demagnetized using eight to ten heating steps. A small secondary component directed along the present field is observed in most flows. This component, which is entirely destroyed between 150°C to 300°C (Fig. 7), is probably a viscous remanent magnetization (VRM) acquired during the last 0.78 Ma. Beyond 300°C a perfect stability of the magnetization direction is

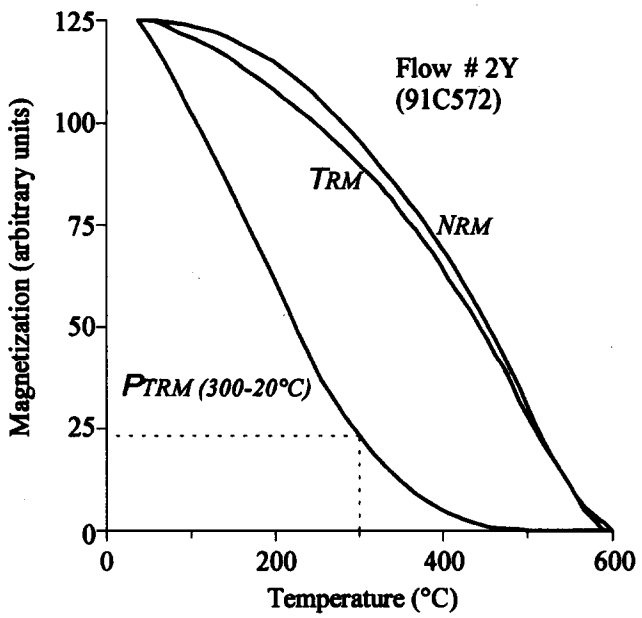


Fig. 6. A representative example of continuous thermomagnetic curves observed for samples of the Thoki basalt. NRM, natural remanent magnetization; TRM, total thermal remanent magnetization acquired, after the first heating, in a 50 μT field; PTRM(300°–20°C), partial thermal remanent magnetization acquired between 300° and 20°C. This PTRM must be completely destroyed at 300°C (temperature of its creation) if the grains are single-domain. The curve obtained here is indicative of a mixture of single/pseudo-single-domain grains.

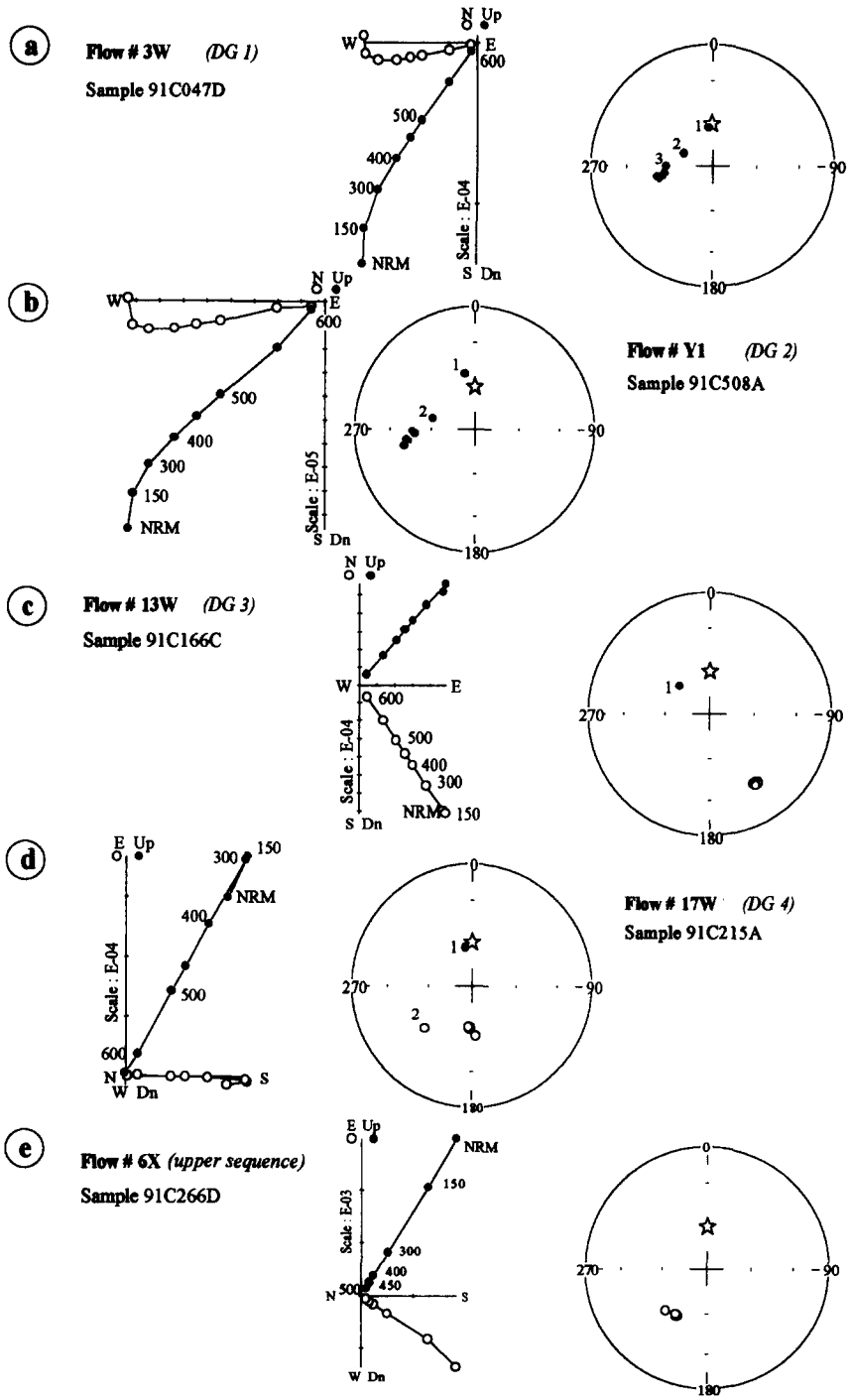


Fig. 7. Example of orthogonal diagrams of progressive thermal demagnetizations (left) and equal-area diagram of vector difference (right) for pilot samples from each directional group. On the orthogonal diagrams, filled circles correspond to the projection onto the vertical plane while open symbols are projection onto the horizontal plane. Unit of NRM intensity is Am^2kg^{-1}. On the equal area projection, filled and open circles represent positive and negative inclinations, respectively. The star is the normal axial dipole field direction.

observed for all the samples and flows (Fig. 7). Given the straightforward behavior of the pilot samples, we chose to use only three heating steps for the remaining samples. For each sample the primary direction was calculated by the least squares method (Kirschvink, 1980). The directions thus obtained were averaged by flow and the statistical parameters were calculated assuming a Fisherian distribution.

Remanent magnetization was measured using a CTF cryogenic magnetometer. Thermal treatment was carried out in field free space using an electrical non-inductive oven manufactured by Pyrox. The samples were left for 30 min at the assigned temperature step. The magnetic susceptibility was measured using a Bartington MS2 susceptibility meter, before the thermal treatment and after each step so as to detect possible magneto-chemical changes during heatings.

5.2. Results

For the calculation of average direction per flow, we eliminated those samples which gave a direction very different from that of the respective mean and for which the field notes indicated some possible orientation error. Finally, 9.6% of the samples were rejected. The average directions are generally very precisely determined (Fig. 8 and Table 1), nearly all α_{95} being less than 5°. The magnetization direction could not be properly determined for flow 7W. The scatter of NRM directions and the abnormally high intensities suggested the presence of isothermal remanent magnetizations (IRM), probably induced by lightning. So this flow was submitted to treatment by alternative field (AF) which resulted in a large reduction of the remanence intensity even for weak AF values. This observation confirms the presence of IRMs. However this treatment did not make it possible to isolate the primary component, except maybe for one sample. Outcrop Y1, which corresponds probably to the same flow, provided a precise average direction close to that of this sample.

An obvious characteristic of the evolution of the average magnetization with elevation along the lower Akhalkalaki sequence (W, Y and Z sections) is the presence of consecutive flows showing the same direction (Fig. 8). Thus, we defined successive directional groups (DGs), each of them corresponding to a set of consecutive lava flows whose α_{95} confidence angles overlap and whose average directions do not

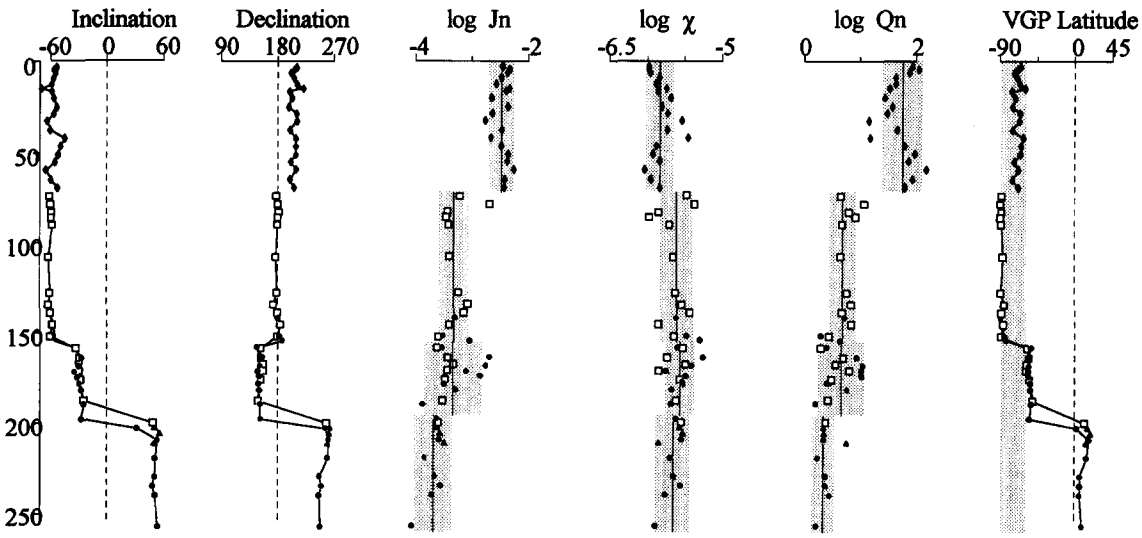


Fig. 8. Evolution of different magnetic parameters for each lava flow of section W (solid circles), section Z (solid triangles), section Y (open squares) and section X (solid diamonds) against the calculated cumulative thickness measured down from the top. See Table 1 for units. Shading denotes range of the standard deviation about each mean except for VGP latitude where shading denotes VGP latitudes > |60°|.

Table 1
Cleaned average directions of magnetization of the lower (sections W, Y, Z) and upper (section X) Akhalkalaki lava flows

Flow	DG	n/N	κ	α_{95}	Inc.	Dec.	Lat.	Long.	J_{nrm}	χ	Q_n
Section W											
1W	1	10/10	131	4.2	52.9	249.1	8.2	351.4	1.30	1.28	1.7
2W	1	7/8	523	2.6	50.2	247.0	5.1	351.0	5.10	1.76	2.9
3W	1	9/10	139	4.4	47.5	251.0	5.9	346.8	8.29	2.87	2.5
4W	1	9/10	207	3.6	50.1	247.6	5.5	350.5	2.56	2.28	2.5
5W	2	10/10	255	3.0	50.5	260.6	13.9	343.1	1.51	2.08	1.8
6W	2	7/7	596	2.5	52.8	262.2	16.2	343.9	2.80	2.95	2.3
7W	2	1/8	N.d.	N.d.	(31.1)	(256.9)	(1.6)	(334.5)	N.d.	N.d.	N.d.
8W	3	2/4	N.d.	N.d.	−28.4	153.0	−54.9	92.9	3.74	2.45	N.d.
9W	3	10/10	238	3.1	−25.4	152.3	−53.0	92.1	1.20	2.11	1.6
10W	3	9/9	246	3.3	−28.1	151.3	−53.6	95.0	6.65	2.16	6.0
11W	3	9/10	891	1.7	−30.5	149.9	−54.0	98.3	4.31	3.05	2.7
12W	3	10/10	462	2.2	−32.8	151.5	−56.1	97.9	22.22	3.32	11.0
13W	3	6/6	498	3.0	−35.5	149.2	−56.0	102.9	8.70	1.85	11.1
14W	3	10/10	488	2.2	−31.0	152.2	−55.6	95.7	31.62	3.95	11.5
15W	3	9/10	209	3.6	−30.5	155.2	−57.1	91.2	25.88	5.74	9.0
16W	3	8/8	139	4.7	−32.8	147.0	−53.2	103.4	2.70	2.59	2.6
17W	4	7/8	225	4.0	−57.1	187.8	−83.0	342.8	11.62	5.07	4.6
18W	4	10/10	191	3.5	−56.3	182.8	−84.9	17.6	3.70	3.45	2.1
19W	4	10/10	201	3.4	−59.9	180.3	−89.3	25.9	5.56	2.45	5.2
Section Y											
1Y	2	10/10	271	2.9	48.0	258.5	11.0	342.6	3.44	2.88	2.5
2Y	3	10/10	562	2.0	−25.8	149.2	−51.4	96.2	2.97	2.47	2.8
3Y	3	4/6	1470	2.4	−29.3	153.4	−55.5	92.9	3.43	2.73	3.2
4Y	3	5/6	499	3.4	−30.9	157.2	−58.4	88.5	7.78	1.45	6.6
5Y	3	5/5	785	2.7	−30.7	157.3	−58.3	88.2	6.39	3.29	3.7
6Y	3	4/5	3363	1.6	−30.3	154.2	−56.4	92.6	4.80	1.85	5.2
7Y	3	5/5	309	4.4	−34.0	153.5	−57.8	96.1	2.56	3.05	2.0
8Y	4	5/5	2035	1.7	−61.5	179.8	−88.9	216.5	2.83	2.29	2.9
9Y	4	5/5	398	3.8	−59.1	184.4	−86.3	338.0	4.03	1.38	7.2
10Y	4	5/5	1079	2.3	−61.5	179.6	−88.6	208.1	8.40	3.75	4.9
11Y	4	4/5	69	11.1	−62.8	173.9	−84.8	167.2	12.21	2.98	7.4
12Y	4	7/8	548	2.6	−61.6	179.5	−88.7	205.5	11.56	2.41	6.2
13Y	4	10/10	301	2.8	−62.8	177.3	−86.6	188.6	4.87	2.26	4.7
14Y	4	7/9	1101	1.8	−59.3	179.8	−88.6	50.6	4.19	1.99	5.0
15Y	4	7/8	820	2.1	−60.3	180.6	−89.5	342.8	3.70	1.07	8.5
16Y	4	5/6	1046	2.4	−60.0	182.5	−88.1	330.3	4.06	1.40	6.8
17Y	4	4/4	698	3.5	−61.0	180.8	−89.2	269.7	24.35	4.34	12.3
18Y	4	5/9	1163	2.2	−62.1	178.3	−87.8	189.3	32.27	3.38	4.6
Section Z											
1Z	2	6/6	448	3.2	49.3	259.7	12.6	342.8	3.55	1.42	5.8
2Z	2	10/10	329	2.7	55.0	262.4	17.7	345.6	3.81	3.04	2.3
3Z	2	10/10	397	2.4	49.8	262.1	14.4	341.7	2.29	2.73	2.3
Section X											
2X	N.d.	5/5	313	4.3	−53.9	205.9	−68.5	324.1	38.43	1.48	65.2
3X	N.d.	10/10	197	3.4	−60.7	199.4	−75.6	306.1	40.68	1.10	88.0
4X	N.d.	3/3	N.d.	N.d.	−65.8	209.4	−68.3	285.8	56.90	0.93	153.4
5X	N.d.	5/5	437	3.7	−56.6	200.7	−73.5	321.8	45.04	1.46	74.3
6X	N.d.	5/5	246	4.9	−53.0	208.4	−66.3	323.6	45.24	1.18	94.6
7X	N.d.	7/7	210	4.2	−50.9	208.4	−65.3	327.9	34.28	1.32	64.4

Table 1 (continued)

Flow	DG	n/N	κ	α_{95}	Inc.	Dec.	Lat.	Long.	J_{nrm}	χ	Q_n
Section X											
9X	N.d.	4/4	597	3.8	−61.7	199.2	−75.7	301.3	36.07	1.87	47.1
10X	N.d.	3/4	N.d.	N.d.	−64.6	210.6	−67.6	290.3	18.00	2.95	15.0
11X	N.d.	6/6	101	6.7	−57.7	210.2	−66.7	311.1	23.83	1.88	31.0
12X	N.d.	4/7	91	9.7	−54.4	196.8	−75.3	334.3	58.07	1.56	38.9
13X	N.d.	6/6	581	2.8	−57.8	202.6	−72.5	315.9	23.50	2.08	27.5
14X	N.d.	1/3	N.d.	N.d.	−60.0	198.0	−76.5	310.0	43.87	N.d.	N.d.
15X	N.d.	3/5	N.d.	N.d.	−70.4	220.8	−60.3	273.1	65.12	1.80	34.0
16X	N.d.	5/5	115	7.2	−60.1	210.5	−67.1	304.4	29.98	1.29	43.8
17X	N.d.	6/7	134	7.8	−58.0	205.8	−70.1	313.1	35.16	1.45	44.5
18X	N.d.	4/5	342	5.0	−55.8	200.4	−73.4	324.9	44.76	1.08	75.6
19X	N.d.	5/5	671	3.0	−57.7	206.4	−69.6	313.4	47.64	1.05	113.0
20X	N.d.	5/5	200	5.4	−54.7	209.9	−65.9	318.7	36.43	1.05	85.2

DG, directional group number; n/N, number of samples used in the analysis/total number of samples collected; κ, precision parameter of Fisher distribution; $\alpha 95$, confidence cone about mean direction; Inc., inclination positive downward; Dec., declination east of north; Lat./Long.; latitude/longitude of VGP position; J_{nrm}, geometric mean intensity before magnetic cleaning ($\times 10^{-4}$ $Am^2.kg^{-1}$); χ, geometric mean susceptibility ($\times 10^{-6}$ SI); Q, geometric mean Koenigsberger ratio.

exhibit a regular trend. Four DGs, were identified (Table 2). As a result, flows cannot be correlated individually between sections upon the basis of paleomagnetic data alone. In contrast, the lava flows from the upper Akhalkalaki sequence (section X) yields in general significantly different directions between consecutive flows. Thus no obvious DG can be defined. Moreover these flows show some magnetic characteristics different from those of the lower Akhalkalaki sequence: average magnetization intensity is higher while magnetic susceptibility is lower (Table 1), which suggests that the magnetic carriers are more abundant or of a smaller grain size. Thus, the stratigraphic discontinuity observed corresponds also to some petrological changes.

Altogether some 16 consecutive flows from the lower Akhalkalaki sequence recorded intermediate field directions. Unfortunately, this record does not provide a detailed record of the evolution of the geomagnetic field since only three distinct intermediate directions could be recognized, corresponding to DG1, DG2 and DG3 (Fig. 9). This figure clearly shows that only the end of either a transition or an excursion was recorded. The initial polarity is not known.

6. Paleointensity determinations

6.1. Experimental procedure

Two variants of the Thellier and Thellier (1959) method of determination of paleointensities were used. Most of the experiments use a simplified version (without PTRM checks) of the standard method. Thus, the double heating steps were performed under a laboratory field of opposite sense and the remanences measured at room temperature. For a few

Table 2
Mean paleomagnetic data for the lower Akhalkalaki sequence directional groups

DG	N_{flow}	N	κ	α_{95}	Inc.	Dec.	δ	Lat.	Long.
1	4	35	168	1.9	50.3	248.8	56.2	6.3	349.9
2	7	54	209	1.3	50.6	260.8	51.5	14.0	343.0
3	15	106	214	0.9	−30.2	152.0	144.4	−55.1	95.5
4	14	91	271	0.9	−60.2	180.8	179.5	−89.3	344.5

See Table 1 for which lavas are in each DG and see Table 1 footnotes for explanation. δ reversal angle from the normal dipole direction.

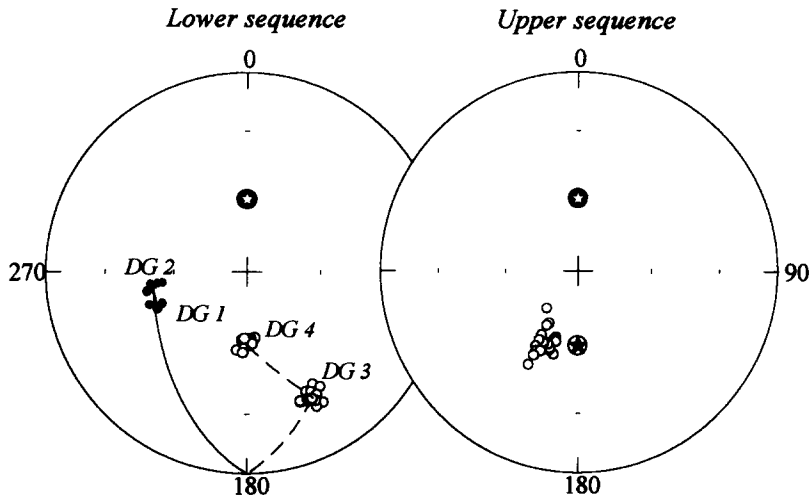


Fig. 9. Equal area projections of the average direction for each lava flow.

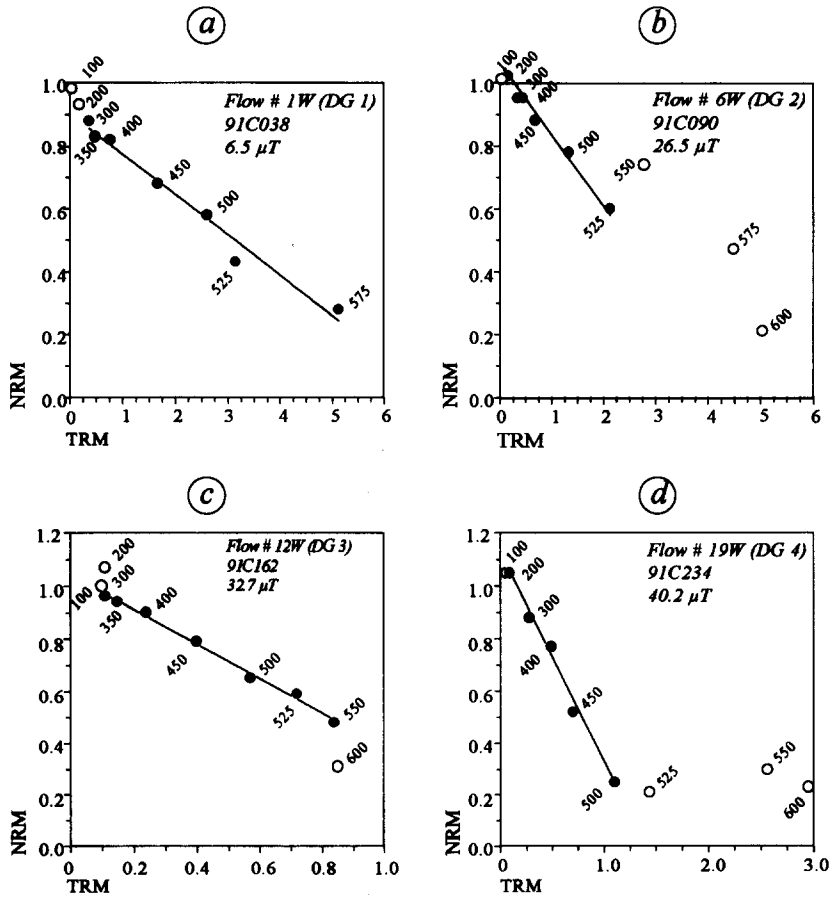


Fig. 10. NRM–TRM diagrams for one specimen by directional group. Open circles are rejected points and solid circles are points used to calculate the least squares line.

Table 3
Thoki basalt paleointensity determinations

DG	Lava flow	Specimen	N	T_{min}–T_{max}	f	g	q	w	Fe ± σ Fe
Section W									
1	1W	91C021	6	400–575	0.606	0.297	0.763	0.381	9.9 ± 2.3
		91C038	7	300–575	0.672	0.773	7.575	3.387	**6.5 ± 0.4**
1	2W	91C043	5	200–450	0.669	0.580	3.643	2.103	11.6 ± 1.2
2	5W	91C075	6	200–525	0.673	0.626	7.058	3.156	**7.8 ± 1.5**
		91C084	5	200–450	0.349	0.649	1.834	1.059	24.5 ± 3.0
2	6W	91C090	6	200–525	0.507	0.559	0.551	0.246	26.5 ± 13.6
3	10W	91C133	6	300–575	0.753	0.749	3.055	1.247	18.6 ± 3.4
3	11W	91C144	9	300–600	0.459	0.820	3.980	1.500	20.6 ± 1.9
		91C145	7	200–525	0.497	0.749	4.871	2.178	15.6 ± 1.2
		91C147	6	300–525	0.320	0.697	2.837	1.418	11.3 ± 0.9
3	12W	91C157	6	300–525	0.351	0.779	2.476	1.238	61.2 ± 6.7
		91C162	7	300–550	0.463	0.804	9.966	4.457	**32.7 ± 1.2**
3	13W	91C167	7	200–525	0.348	0.786	2.005	0.896	24.2 ± 3.3
		91C169	8	300–575	0.466	0.836	7.277	2.970	**45.0 ± 2.4**
3	14W	91C173	7	200–525	0.360	0.792	1.807	0.808	40.5 ± 6.4
3	16W	91C202	7	200–525	0.600	0.785	2.295	1.026	34.1 ± 7.0
4	17W	91C215	5	350–525	0.529	0.727	4.290	2.477	16.5 ± 1.5
		91C216	7	200–525	0.372	0.798	3.541	1.584	98.1 ± 8.2
4	18W	91C225	5	200–525	0.689	0.703	13.739	7.932	**23.7 ± 0.8**
4	19W	91C234	5	200–500	0.718	0.728	9.312	5.376	**40.2 ± 2.2**
		91C404	7	200–525	0.422	0.709	1.068	0.478	42.0 ± 11.8
Section Y									
2	1Y	91C518	5	200–450	0.348	0.729	2.035	1.175	48.5 ± 6.1
3	2Y	91C547	6	200–500	0.468	0.610	1.866	0.933	19.9 ± 3.0
		91C572	5	200–450	0.285	0.637	0.615	0.355	35.6 ± 10.5
3	4Y	91C610 *	7	200–500	–	–	–	–	23.0 ± 9.5
		91C607	10	200–600	0.712	0.871	6.372	2.253	**17.0 ± 1.7**
3	6Y	91C620	7	200–525	0.357	0.816	2.346	1.049	45.2 ± 5.6
3	7Y	91C314	6	200–500	0.336	0.759	2.708	1.354	30.2 ± 2.8
4	9Y	91C321	6	200–500	0.275	0.788	3.468	1.734	65.1 ± 4.1
		91C325	7	200–525	0.357	0.782	1.758	0.786	67.3 ± 10.7
4	10Y	91C328 *	6	200–550	–	–	–	–	50.3
4	11Y	91C332 *	5	200–575	–	–	–	–	47.8
		91C332	7	200–525	0.278	0.664	1.239	0.554	80.4 ± 12.0
		91C333	7	200–525	0.451	0.596	4.226	1.890	57.0 ± 3.6
		91C335	7	300–525	0.513	0.509	7.426	3.321	**25.3 ± 0.9**
4	12Y	91C339	5	200–450	0.364	0.503	1.518	0.877	46.6 ± 5.6
4	15Y	91C361	10	200–600	0.890	0.656	7.923	2.801	**23.9 ± 1.8**
		91C363	9	200–575	0.839	0.806	33.698	12.736	**17.2 ± 0.3**
4	16Y	91C369	7	300–575	0.749	0.671	10.740	4.803	**27.7 ± 1.3**
		91C372	8	300–600	0.638	0.753	6.374	2.602	**27.5 ± 2.1**
		91C374	4	200–400	0.285	0.655	3.473	2.456	54.7 ± 2.9
4	17Y	91C377	5	200–500	0.404	0.717	13.228	7.637	**91.3 ± 2.0**
		91C380	5	350–525	0.307	0.660	1.381	0.797	149.8 ± 22.0
Section X									
N.d.	8X	91C444 *	5	100–300	–	–	–	–	52.5

DG, directional group number, * modified Thellier method; N, number of points in the T_{min}–T_{max} interval; T_{min}–T_{max}, temperature interval used to determine paleointensity; f, NRM fraction; g, gap factor; q, quality factor; w, weighting factor (for f, g, q, w, see Coe et al., 1978); Fe, paleointensity estimate for individual sample in μT; σ Fe, standard error; boldfaced fonts indicate the reliable paleointensity determinations ($q \geq 5$).

samples, we used another version in which the remaining NRM and the laboratory TRM intensities are measured at temperature, the double heating steps being performed first in zero field then under the laboratory field (Day et al., 1977). For this second method, the experimental procedure consists of heating the sample in zero field from T_n to T_{n+1}, at which temperature the remanent magnetization J_1 is measured. The sample is then cooled (always in zero field) down to a temperature T_n and the remanent magnetization J_2 measured. The second cycle consists of heating the sample again to T_{n+1} and then cooling it back in a controlled field ($H = H_{lab}$) to a temperature T_n where the remanent magnetization J_3 is measured. This procedure is repeated for successive increasing temperature intervals. The paleointensity is then determined from a plot, at each temperature, of the sum of difference in NRM ($\Sigma\{J_1-J_2\}$) against the sum of the consecutive differences in TRM ($\Sigma\{J_3-J_2\}$). Thus the only difference with the usual Arai–Nagata plot is that the contribution of the highest temperature intervals to the paleointensity determination is reduced as a result of the decay of spontaneous magnetization when T increases. For both methods, the NRM–TRM diagrams were statistically analyzed as recommended by Coe et al. (1978) following Prévot et al.'s (1985) modifications.

All samples studied but one belong to the lower Akhalkalaki sequence. Due to practical problems for transferring the specimens from one laboratory to the others, the rock magnetic and the paleointensity experiments had to be carried out on a collection consisting of only one specimen per flow. Thus magnetic properties (except viscosity index) and paleointensity experiments were performed on different cores. As a result, no pre-selection of the cores was made before undertaking paleointensity experiments.

6.2. Results

Criteria for rejections of results were applied a posteriori. Only the paleointensities determined from a least five point segments and spanning at least 20% of the total NRM were accepted. Fig. 10 shows an example of NRM–TRM diagram obtained by the standard paleointensity procedure for each directional group.

In spite of the rather encouraging indications provided by the rock magnetic experiments reported above, the quality of the paleointensity data is somewhat poor. A paleointensity value was calculated for 46 samples (Table 3). However, the quality factor (which can sometimes reach 100 or more, Prévot et al., 1983) is low and not much confidence can be given to many of these individual data. We did not calculate flow averages because of the small number of determinations per flow (we often have only one determination) and the scattering observed in several cases within single flows. It is reasonable to consider that only paleointensity data corresponding to quality factors equal or greater than 5 can be reliable (bold font in Table 3). Thus, we have indications that the field intensity was particularly low (less than 10 μT) for the first DG. Then, the field magnitude varied between 20 and 40 μT, which is almost twice less than expected at this latitude for the non-transitional field. This corroborates the results of Borisova and Sologashvili (1985) on the same section. This trend suggests a progressive recovering of a normal intensity during the end of either a reversal or an excursion. The low magnitudes obtained for the intermediate field directions agree also with those reported for other volcanic records studied with the Thellier paleointensity method (Bogue and Coe, 1984; Coe et al. 1984; Prévot et al. 1985; Marshall et al., 1988; Roperch et al., 1988; Chauvin et al., 1990; Levi et al., 1990). The possibility that high intensities occur just after reversal, as suggested in some previous studies of volcanic records (Prévot et al., 1985; Bogue and Paul, 1993), seems to be supported by the determination obtained from lava flow 17Y. However, many more paleointensity measurements are needed to check this suggestion.

7. Discussion

The paleomagnetic results reported here qualitatively corroborate the preliminary data of Borisova and Sologashvili (1985) and Sologashvili (1986) regarding both the direction and paleointensity of the paleofield. However, we found no evidence that these intermediate directions are preceeded by normal flows belonging to the Akhalkalaki sequence. Obviously, the polarity of the underlying Goderzi

formation is of no help in this matter because of the hiatus in volcanic activity which is attested by the erosion surface separating these two formations. Thus, the geomagnetic event recorded in the Thoki section is not necessarily a N–T–R reversal, as previously supposed (Borisova and Sologashvili, 1985). This might also be a R–T–R excursion. Also, the age of these intermediate directions is quite different from the Gauss–Matuyama age assumed by Borisova and Sologashvili (1985).

The identification of the geomagnetic event recorded in the lower Akhalkalaki sequence near Thoki is uncertain. If one refers to the polarity time scale of Cande and Kent (1992), the age reported here (3.60 ± 0.60 Ma) does not correspond to a known reversal but rather to some (unspecified) excursion which occurred during chron 2Ar of the Gilbert epoch. The existence of such an excursion is supported by some paleomagnetic observations in South West Iceland (Kristjansson et al., 1980) indicating the presence of one flow (FA17) with an intermediate magnetization direction within a reversed polarity zone believed to correspond to that chron.

An alternative hypothesis is that the Thoki geomagnetic event corresponds to the end of the Cochiti–Gilbert reversal. This N–R boundary is dated at 4.0 Ma on the Cande and Kent (1992) time scale. Our paleomagnetic data might then be compared to the intermediate directions reported by Watkins et al. (1977) for the NP section in the Borgarfjördur area (West Iceland) or/and to those found in the S and T sections of East Iceland (Watkins and Walker, 1977).

Another point of interest is the origin of the VGP clusters observed during this transition (Fig. 11). Groups of consecutive flows with identical paleofield directions are commonly observed in volcanic records of geomagnetic reversals. In their detailed study of the Steens Mountain reversal, Mankinen et al. (1985) interpreted these groups as being due to the eruption of successive lava flows in a very short time interval. In contrast, Hoffman (1992) assumed that these clusters indicate long lived transitional states of the geomagnetic field. He also suggested that such stages are dominantly dipolar and have been persistent features of the geomagnetic reversals for the past 10 Ma. The question of whether the VGP clusters are of volcanic or geomagnetic origin is still pending. What is clear, however, is that our VGP clusters do not match those of Hoffman (Fig. 11) thus, such clusters are not witnesses of a mainly dipolar field recurrently present during successive reversals and keeping the same orientation.

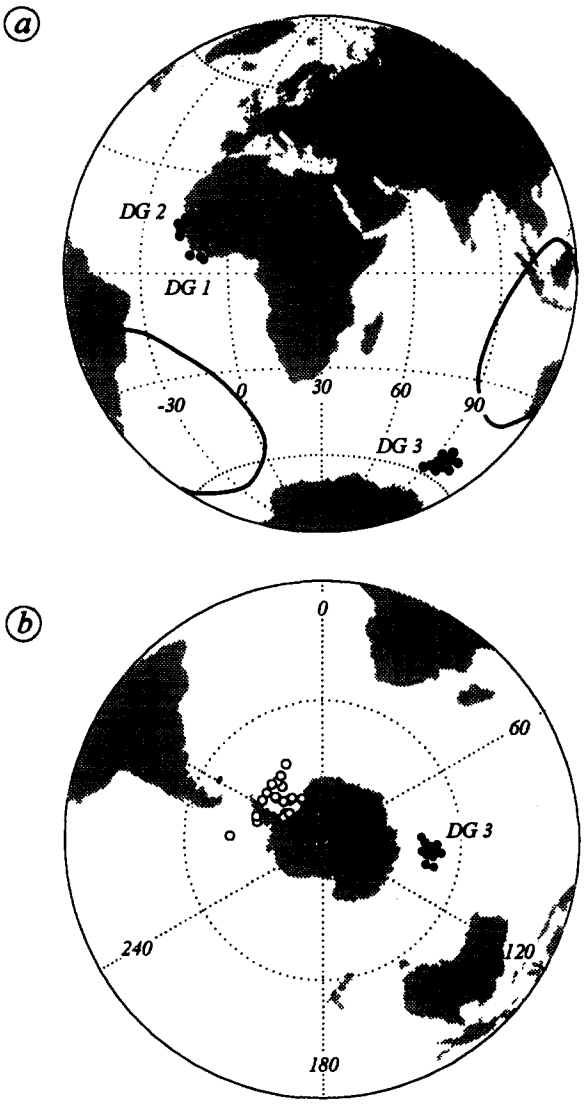


Fig. 11. Projection of the VGPs corresponding to the directions shown on Fig. 9. Filled circles are VGPs measured along the lower Akhalkalaki sequence while open circles are those measured along the upper Akhalkalaki sequence.

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